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Differential gear

Abstract

A differential gear includes a first connecting member with straight-spline protrusions on its inner periphery; a second connecting member with straight-spline protrusions on its inner periphery; a first side gear; and a second side gear. The first side gear includes a first cylindrical portion coupled to the first connecting member by helical splines, and a first wall portion extending radially inward. The second side gear includes a second cylindrical portion coupled to the second connecting member by helical splines, and a second wall portion extending radially inward. The first wall portion of the first side gear protrudes radially inward beyond the straight-spline protrusions of the first connecting member, and the second wall portion of the second side gear protrudes radially inward beyond the straight-spline protrusions of the second connecting member.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to differential gears for vehicles that distribute the driving force of a driving source to a pair of output shafts while allowing differential rotation between the output shafts.

BACKGROUND ART

- (2) Some differential gears that distribute the driving force of a driving source for a vehicle to a pair of output shafts while allowing differential rotation between the output shafts have been known to have a limited slip function to limit differential rotation between the output shafts. For example, a vehicle equipped with such a differential gear can transmit the driving force to a left wheel even when a right wheel slips. Travel stability is thus improved. The applicant of the present disclosure has proposed the differential gears with the limited slip function described in PTL 1, 2.
- (3) The differential gear described in PTL 1 includes: a first side gear and a second side gear that are disposed side by side in an axial direction; a plurality of pinion gear sets each composed of a first pinion gear meshing with the first side gear and a second pinion gear meshing with the second side gear; and a differential case holding the plurality of pinion gear sets. The first and second pinion gears and the first and second side gears are helical gears that generate an axial thrust force by meshing with each other. When torque is transferred from the differential case to a pair of output shafts, end faces of the first and second side gears are pressed against washers disposed in the differential case due to the thrust forces, generating frictional forces that reduce differential rotation between the pair of output shafts.
- (4) The differential gear described in PTL 2 includes, in addition to the above configuration, a first connecting member coupled to the first side gear by helical splines and a second connecting member coupled to the second side gear by helical spines in order to further increase differential limiting forces. When torque is transferred from the differential case to the pair of output shafts, the differential limiting forces increase due to the thrust forces additionally generated by the helical spline couplings.
- (5) In the differential gear described in PTL 1, an axle spacer is disposed between the first side gear and the second side gear. When assembling a vehicle, right and left axle shafts that are the pair of output shafts to be inserted through the first side gear and the second side gear, respectively, are caused to abut on both end faces of the axle spacer. The right and left axle shafts are thus positioned, which facilitates assembly of the vehicle. The differential gear described in PTL 2 does not include such an axle spacer.

CITATION LIST

Patent Literature

(6) PTL 1: JP 2015-203483 A

(7) PTL 2: JP 2021-173389 A

SUMMARY OF INVENTION

Technical Problem

(8) The axle spacer of the differential gear described in PTL 1 contributes to facilitate assembly of the vehicle as described above. Once the vehicle is assembled, however, the axle spacer no longer performs any function, but rather obstructs the flow of lubricating oil in the differential case. The differential gear described in PTL 2 does not include an axle spacer. This can reduce the number of parts of the differential gear and the assembly time. However, the pair of output shafts (right and left axle shafts) is unable to be positioned by abutting on the axle spacer when assembling a vehicle.

(9) The present invention was made in view of the above circumstances, and it is an object of the present invention to provide a differential gear that facilitates positioning of a pair of output shafts without using an axle spacer when assembling a vehicle.

Solution to Problem

(10) In order to achieve the above object, the present invention provides a differential gear that distributes a driving force for a vehicle to a first output shaft and a second output shaft. The differential gear includes: a cylindrical first connecting member including, on an inner periphery of the first connecting member, a plurality of straight-spline protrusions to which the first output shaft is coupled by straight splines; a cylindrical second connecting member including, on an inner periphery of the second connecting member, a plurality of straight-spline protrusions to which the

second output shaft is coupled by straight splines; a first side gear combined with the first connecting member; a second side gear combined with the second connecting member; a plurality of pinion gear sets each composed of a first pinion gear and a second pinion gear being meshed with each other, the first pinion gear meshing with the first side gear, and the second pinion gear meshing with the second side gear; a differential case that holds the plurality of pinion gear sets; and a center washer that is disposed between the first side gear and the second side gear, rotation of the center washer relative to the differential case being restricted. The first side gear includes: a first cylindrical portion disposed around an outer periphery of the first connecting member and coupled to the first connecting member by helical splines; and a first wall portion extending radially inward from one end on the center washer side of the first cylindrical portion, the first wall portion protruding radially inward beyond the plurality of straight-spline protrusions of the first connecting member. The second side gear includes: a second cylindrical portion disposed around an outer periphery of the second connecting member and coupled to the second connecting member by helical splines; and a second wall portion extending radially inward from one end on the center washer side of the second cylindrical portion, the second wall portion protruding radially inward beyond the plurality of straight-spline protrusions of the second connecting member. During driving in which torque is transferred from the first side gear and the second side gear to the first connecting member and the second connecting member, the center washer generates a frictional force between the center washer and each of the first wall portion of the first side gear and the second wall portion of the second side gear by a thrust force generated by coupling by the helical splines.

Advantageous Effects of Invention

(11) The differential gear according to the present invention facilitates axial positioning of the first output shaft and the second output shaft without using an axle spacer when assembling a vehicle.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a sectional view showing an example of the configuration of a differential gear according to an embodiment of the present invention.
- (2) FIG. 2 is an exploded perspective view of the differential gear.
- (3) FIG. 3 illustrates a first connecting member, a second connecting member, a first side gear, a second side gear, a center washer, a first side washer, and a second side washer shown spaced apart from each other in an axial direction.
- (4) FIG. 4 is a sectional view showing an example of the configuration of a differential gear according to a comparative example.

DESCRIPTION OF EMBODIMENTS

Embodiment

(5) An embodiment of the present invention will be described with reference to FIGS. 1 to 3. The embodiment described below is shown as a suitable specific example for carrying out the present invention, and part of the embodiment specifically illustrates various technically preferable technical matters. However, the technical scope of the present invention is not limited to such specific aspects.

(6) FIG. 1 is a sectional view showing an example of the configuration of a differential gear according to the embodiment of the present invention. FIG. 2 is an exploded perspective view of the differential gear.

(7) A differential gear 1 is mounted on a vehicle and is used to distribute the driving force of a driving source to a first output shaft 91 and a second output shaft 92 while allowing differential rotation between these output shafts. In FIG. 1, the first output shaft 91 and the second output shaft

92 are shown by hidden outlines. The driving source is, for example, an engine that is an internal combustion engine, an electric motor, or a combination of an engine and an electric motor.

(8) The differential gear **1** includes: a cylindrical first connecting member **2** that rotates with the first output shaft **91**; a cylindrical second connecting member **3** that rotates with the second output shaft **92**; a first side gear **4** combined with the first connecting member **2**; a second side gear **5** combined with the second connecting member **3**; a plurality of pinion gear sets **6**; a differential case **7** holding the plurality of pinion gear sets **6**; and a center washer **80**, a first side washer **81**, and a second side washer **82** that are held in the differential case **7** so as not to be rotatable relative to the differential case **7**.

(9) A ring gear **11** shown by hidden outlines in FIG. **1** is fixed to the differential case **7** with a plurality of bolts **12**. The differential case **7** is rotated about a rotation axis **O** by the driving force of the driving source applied from the ring gear **11**. The first connecting member **2**, the second connecting member **3**, the first side gear **4**, and the second side gear **5** are rotatable relative to the differential case **7** about the rotation axis **O**. Hereinafter, a direction parallel to the rotation axis **O** will be referred to as axial direction. A direction perpendicular to the rotation axis **O** will be referred to as radial direction.

(10) The differential case **7** includes a bottomed cylindrical case body **71** and a case lid **72** disposed so as to close an opening of the case body **71**. The case body **71** and the case lid **72** are coupled with a plurality of bolts **73**. The case body **71** is a single-piece member composed of: a cylindrical tube portion **711**; a disc-shaped first sidewall portion **712** formed so as to extend radially inward from an end on one side in the axial direction of the tube portion **711**; a cylindrical first extended portion **713** extended in the axial direction from the central portion of the first sidewall portion **712**; and a first flange portion **714** formed so as to protrude radially outward from an end on the other side in the axial direction of the tube portion **711**. An oil hole **712a** for passing lubricating oil therethrough is formed in the first sidewall portion **712**.

(11) The case lid **72** is a single-piece member composed of: a disc-shaped second sidewall portion **721**; a cylindrical second extended portion **722** extended in the axial direction from the central portion of the second sidewall portion **721**; and a second flange portion **723** formed so as to protrude radially outward from the second sidewall portion **721**. An oil hole **721a** for passing lubricating oil therethrough is formed in the second sidewall portion **721**.

(12) The bolts **12** for fixing the ring gear **11** are inserted through the first flange portion **714** of the case body **71** and the second flange portion **723** of the case lid **72**. The differential case **7** and the ring gear **11** are housed in a differential carrier, not shown, and the ring gear **11** is meshed with a pinion gear in the differential carrier. Bearings, not shown, are mounted around the first extended portion **713** and the second extended portion **722**, and the differential case **7** is supported by the bearings so as to be rotatable relative to the differential carrier. Lubricating oil (differential oil) is sealed in the differential carrier.

(13) The center washer **80** is held in the tube portion **711** of the case body **71**. The first side washer **81** is held by the first sidewall portion **712** of the case body **71**, and the second side washer **82** is held by the second sidewall portion **721** of the case lid **72**. The center washer **80** is disposed between the first side gear **4** and the second side gear **5**. The first side washer **81** is disposed at such a position that the first connecting member **2** and the first side gear **4** are sandwiched between the first side washer **81** and the center washer **80**. The second side washer **82** is disposed at such a position that the second connecting member **3** and the second side gear **5** are sandwiched between the second side washer **82** and the center washer **80**.

(14) As shown in FIG. **2**, each of the center washer **80**, the first side washer **81**, and the second side washer **82** has a disc-shaped disc portion **801**, **811**, **821** and a plurality of protruding portions **802**, **812**, **822** protruding radially outward from the outer edge of the disc-shaped disc portion **801**, **811**, **821**. Rotation of the center washer **80** relative to the differential case **7** is restricted by the plurality of protruding portions **802** engaging with the tube portion **711** of the case body **71**. Rotation of the

first side washer **81** relative to the differential case **7** is restricted by the plurality of protruding portions **812** engaging with the first sidewall portion **712** of the case body **71**. Rotation of the second side washer **82** relative to the differential case **7** is restricted by the plurality of protruding portions **822** engaging with the second sidewall portion **721** of the case lid **72**.

(15) In the present embodiment, four pinion gear sets **6** are housed in the differential case **7**. Each pinion gear set **6** is composed of one first pinion gear **61** and two second pinion gears **62** meshing with each other. The first pinion gear **61** has a large diameter gear portion **611** and a small diameter gear portion **612** that are different in pitch circle diameter. The two second pinion gears **62** have a shorter axial length than the first pinion gear **61**, and are meshed with the small diameter gear portion **612** of the first pinion gear **61**.

(16) Four bores **70** for holding the four pinion gear sets **6** are formed in the case body **71**. The inner surface of each bore **70** includes a sliding surface **70a** on which the tooth tips of the first pinion gear **61** slide, and sliding surfaces **70b** on which the tooth tips of the second pinion gears **62** slide. When the vehicle travels straight, the first pinion gear **61** and the second pinion gears **62** rotate (revolve) with the differential case **7** without rotating about their axes in the bore **70**. When the first output shaft **91** and the second output shaft **92** rotate differentially as the vehicle makes a turn etc., the first pinion gear **61** and the second pinion gears **62** rotate with the differential case **7** while rotating about their axes in the bore **70**.

(17) FIG. **3** illustrates the first connecting member **2**, the second connecting member **3**, the first side gear **4**, the second side gear **5**, the center washer **80**, the first side washer **81**, and the second side washer **82** shown spaced apart from each other in the axial direction.

(18) The first connecting member **2** is a single-piece member composed of: a first tubular base portion **21** formed in a cylindrical shape; an annular first outer peripheral wall portion **22** formed so as to protrude radially outward from an end on the first side washer **81** side of the first tubular base portion **21**; and an annular first support wall portion **23** formed adjacent to the first outer peripheral wall portion **22** so as to protrude radially outward from the first tubular base portion **21**. A straight-spline fitting portion **212** having a plurality of straight-spline protrusions **211** extending in the axial direction is formed on the inner periphery of the first tubular base portion **21**. A helical-spline fitting portion **214** having a plurality of helical-spline protrusions **213** tilted with respect to the axial direction is formed on the outer periphery of the first tubular base portion **21**. An annular recess **20** for machining the plurality of helical-spline protrusions **213** is formed between the helical-spline fitting portion **214** and the first support wall portion **23**.

(19) Similarly, the second connecting member **3** is a single-piece member composed of: a second tubular base portion **31** formed in a cylindrical shape; an annular second outer peripheral wall portion **32** protruding radially outward from an end on the second side washer **82** side of the second tubular base portion **31**; and an annular second support wall portion **33** formed adjacent to the second outer peripheral wall portion **32** so as to protrude radially outward from the second tubular base portion **31**. A straight-spline fitting portion **312** having a plurality of straight-spline protrusions **311** extending in the axial direction is formed on the inner periphery of the second tubular base portion **31**. A helical-spline fitting portion **314** having a plurality of helical-spline protrusions **313** tilted with respect to the axial direction is formed on the outer periphery of the second tubular base portion **31**. An annular recess **30** for machining the plurality of helical-spline protrusions **313** is formed between the helical-spline fitting portion **314** and the second support wall portion **33**.

(20) The first output shaft **91** is inserted into the first tubular base portion **21** and is coupled to the plurality of straight-spline protrusions **211** of the straight-spline fitting portion **212** by straight splines. The second output shaft **92** is inserted into the second tubular base portion **31** and is coupled to the plurality of straight-spline protrusions **311** of the straight-spline fitting portion **312** by straight splines.

(21) The first side gear **4** is a single-piece member composed of: a first cylindrical portion **41**

disposed around the outer periphery of the first connecting member **2** and coupled to the first connecting member **2** by helical splines; and a first wall portion **42** extending radially inward from one end on the center washer **80** side of the first cylindrical portion **41**. A helical-spline fitting portion **412** having a plurality of helical-spline protrusions **411** tilted with respect to the axial direction is formed on the inner periphery of the first cylindrical portion **41**. A helical gear portion **414** having a plurality of helical teeth **413** tilted with respect to the axial direction is formed on the outer periphery of the first cylindrical portion **41**.

(22) Similarly, the second side gear **5** is a single-piece member composed of: a second cylindrical portion **51** disposed around the outer periphery of the second connecting member **3** and coupled to the second connecting member **3** by helical splines; and a second wall portion **52** extending radially inward from one end on the center washer **80** side of the second cylindrical portion **51**. A helical-spline fitting portion **512** having a plurality of helical-spline protrusions **511** tilted with respect to the axial direction is formed on the inner periphery of the second cylindrical portion **51**. A helical gear portion **514** having a plurality of helical teeth **513** tilted with respect to the axial direction is formed on the outer periphery of the second cylindrical portion **51**.

(23) The helical-spline fitting portion **412** of the first side gear **4** is fitted on the helical-spline fitting portion **214** of the first connecting member **2**, and the plurality of helical-spline protrusions **411** of the first side gear **4** and the plurality of helical-spline protrusions **213** of the first connecting member **2** are meshed with each other. The helical gear portion **414** of the first side gear **4** meshes with the large diameter gear portion **611** of the first pinion gear **61**. When torque is transferred from the differential case **7** to the first output shaft **91**, the helical-spline fitting portion **412** and helical gear portion **414** of the first side gear **4** and the helical-spline fitting portion **214** of the first connecting member **2** are subjected to an axial thrust force.

(24) Similarly, the helical-spline fitting portion **512** of the second side gear **5** is fitted on the helical-spline fitting portion **314** of the second connecting member **3**, and the plurality of helical-spline protrusions **511** of the second side gear **5** and the plurality of helical-spline protrusions **313** of the second connecting member **3** are meshed with each other. The helical gear portion **514** of the second side gear **5** meshes with the second pinion gears **62**. When torque is transferred from the differential case **7** to the second output shaft **92**, the helical-spline fitting portion **512** and helical gear portion **514** of the second side gear **5** and the helical-spline fitting portion **314** of the second connecting member **3** are subjected to an axial thrust force.

(25) As shown in FIG. 3, an end face **22a** on the center washer **80** side of the first outer peripheral wall portion **22** of the first connecting member **2** faces an end face **41a** on the opposite side of the first cylindrical portion **41** of the first side gear **4** from the center washer **80** (on the first side washer **81** side) in the axial direction. The end face **22a** of the first outer peripheral wall portion **22** and the end face **41a** of the first cylindrical portion **41** are flat surfaces perpendicular to the axial direction. An inner peripheral surface **41b** of an end on the first side washer **81** side of the first cylindrical portion **41** of the first side gear **4** faces an outer peripheral surface **23a** of the first support wall portion **23** of the first connecting member **2** in the radial direction. With the outer peripheral surface **23a** being in contact with the inner peripheral surface **41b** of the first cylindrical portion **41**, the first support wall portion **23** stabilizes the attitude of the first side gear **4** with respect to the first connecting member **2**.

(26) An end face **32a** on the center washer **80** side of the second outer peripheral wall portion **32** of the second connecting member **3** faces an end face **51a** on the opposite side of the second cylindrical portion **51** of the second side gear **5** from the center washer **80** (on the second side washer **82** side) in the axial direction. The end face **32a** of the second outer peripheral wall portion **32** and the end face **51a** of the second cylindrical portion **51** are flat surfaces perpendicular to the axial direction. An inner peripheral surface **51b** of an end on the second side washer **82** side of the second cylindrical portion **51** of the second side gear **5** faces an outer peripheral surface **33a** of the second support wall portion **33** of the second connecting member **3** in the radial direction. With the

outer peripheral surface **33a** being in contact with the inner peripheral surface **51b** of the second cylindrical portion **51**, the second support wall portion **33** stabilizes the attitude of the second side gear **S** with respect to the second connecting member **3**.

(27) A surface on the first side washer **81** side of the first connecting member **2** includes an abutting surface **2a** that generates a frictional force through frictional contact with the first side washer **81**, and a non-contact surface **2b** that does not contact the first side washer **81**. The abutting surface **2a** and the non-contact surface **2b** are annular flat surfaces perpendicular to the axial direction, and the abutting surface **2a** is formed radially outward of the non-contact surface **2b**. The abutting surface **2a** protrudes toward the first side washer **81** with respect to the non-contact surface **2b**. An end face **21a** on the center washer **80** side of the first tubular base portion **21** of the first connecting member **2** faces the first wall portion **42** of the first side gear **4** in the axial direction.

(28) Similarly, a surface on the second side washer **82** side of the second connecting member **3** includes an abutting surface **3a** that generates a frictional force through frictional contact with the second side washer **82**, and a non-contact surface **3b** that does not contact the second side washer **82**. The abutting surface **3a** and the non-contact surface **3b** are annular flat surfaces perpendicular to the axial direction, and the abutting surface **3a** is formed radially outward of the non-contact surface **3b**. The abutting surface **3a** protrudes toward the second side washer **82** with respect to the non-contact surface **3b**. An end face **31a** on the center washer **80** side of the second tubular base portion **31** of the second connecting member **3** faces the second wall portion **52** of the second side gear **5** in the axial direction.

(29) The first wall portion **42** of the first side gear **4** protrudes radially inward beyond the plurality of straight-spline protrusions **211** of the first connecting member **2**. When assembling a vehicle on which the differential gear **1** is mounted, the first output shaft **91** is inserted into the first extended portion **713** of the differential case **7** and the first connecting member **2** to abut on the first wall portion **42** of the first side gear **4**. A through hole **420** extending through the first wall portion **42** in the axial direction is formed in the central portion of the first side gear **4**. As shown in FIG. 3, D.sub.11 represents the inner diameter of the through hole **420** of the first side gear **4**, and D.sub.12 represents the inner diameter of the straight-spline fitting portion **212** of the first connecting member **2**. The inner diameter D.sub.11 of the through hole **420** is smaller than the inner diameter D.sub.12 of the straight-spline fitting portion **212**.

(30) A surface on the opposite side of the first wall portion **42** of the first side gear **4** from the center washer **80** includes a first opposing surface **42a** that faces the end face **21a** of the first tubular base portion **21** of the first connecting member **2**, and a first abutting surface **42b** on which a distal end face **91a** (see FIG. 1) of the first output shaft **91** abuts. The first abutting surface **42b** is formed on a portion of the first wall portion **42** that protrudes radially inward beyond the plurality of straight-spline protrusions **211** of the first connecting member **2**. The first abutting surface **42b** has been subjected to a finishing process for improving the dimensional accuracy of the first output shaft **91** when the distal end face **91a** of the first output shaft **91** is caused to abut on the first abutting surface **42b**. The first abutting surface **42b** has thus been slightly offset toward the center washer **80** with respect to the first opposing surface **42a**.

(31) Similarly, the second wall portion **52** of the second side gear **5** protrudes radially inward beyond the plurality of straight-spline protrusions **311** of the second connecting member **3**. When assembling the vehicle on which the differential gear **1** is mounted, the second output shaft **92** is inserted into the second extended portion **722** of the differential case **7** and the second connecting member **3** to abut on the second wall portion **52** of the second side gear **5**. A through hole **520** extending through the second wall portion **52** in the axial direction is formed in the central portion of the second side gear **5**. As shown in FIG. 3, D.sub.21 represents the inner diameter of the through hole **520** of the second side gear **5**, and D.sub.22 represents the inner diameter of the straight-spline fitting portion **312** of the second connecting member **3**. The inner diameter D.sub.21

of the through hole **520** is smaller than the inner diameter $D_{sub.22}$ of the straight-spline fitting portion **312**.

(32) A surface on the opposite side of the second wall portion **52** of the second side gear **5** from the center washer **80** includes a second opposing surface **52a** that faces the end face **31a** of the second tubular base portion **31** of the second connecting member **3**, and a second abutting surface **52b** on which a distal end face **92a** (see FIG. **1**) of the second output shaft **92** abuts. The second abutting surface **52b** is formed on a portion of the second wall portion **52** that protrudes radially inward beyond the plurality of straight-spline protrusions **311** of the second connecting member **3**. The second abutting surface **52b** has been subjected to a finishing process for improving the dimensional accuracy of the second output shaft **92** when the distal end face **92a** of the second output shaft **92** is caused to abut on the second abutting surface **52b**. The second abutting surface **52b** has thus been slightly offset toward the center washer **80** with respect to the second opposing surface **52a**.

(33) The first output shaft **91** is temporarily caused to abut on the first wall portion **42** of the first side gear **4** when assembling the vehicle. However, when the assembly is completed, the first output shaft **91** is positioned away from the first wall portion **42** of the first side gear **4** in the axial direction, and does not contact the first wall portion **42**. Similarly, the second output shaft **92** is temporarily caused to abut on the second wall portion **52** of the second side gear **5** when assembling the vehicle. However, when the assembly is completed, the second output shaft **92** is positioned away from the second wall portion **52** of the second side gear **5** in the axial direction, and does not contact the second wall portion **52**.

(34) During driving in which the vehicle accelerates in a forward direction or moves forward at a constant speed, torque is transferred from the first side gear **4** to the first connecting member **2** and from the second side gear **5** to the second connecting member **3**. On the other hand, during coasting in which the vehicle decelerates by engine braking or regenerative braking of an electric motor while traveling forward, torque is transferred from the first connecting member **2** to the first side gear **4** and from the second connecting member **3** to the second side gear **5**. However, the torque that is transferred between the first side gear **4** and the first connecting member **2** and between the second side gear **5** and the second connecting member **3** is relatively smaller during coasting than during driving.

(35) In FIG. **3**, the directions of axial thrust forces $F_{sub.11}$ to $F_{sub.16}$ that are generated on the first connecting member **2**, the second connecting member **3**, the first side gear **4**, and the second side gear **5** during driving are shown by black arrows. In FIG. **3**, the directions of axial thrust forces $F_{sub.21}$ to $F_{sub.26}$ that are generated on the first connecting member **2**, the second connecting member **3**, the first side gear **4**, and the second side gear **5** during coasting are shown by white arrows. During driving, the thrust force $F_{sub.11}$ toward the center washer **80** is generated on the first side gear **4** by the meshing between the helical gear portion **414** of the first side gear **4** and the large diameter gear portion **611** of the first pinion gear **61**, and the thrust force $F_{sub.12}$ toward the center washer **80** is generated on the second side gear **5** by the meshing between the helical gear portion **514** of the second side gear **5** and the second pinion gear **62**.

(36) During driving, the thrust force $F_{sub.13}$ toward the center washer **80** is generated on the first side gear **4** and the thrust force $F_{sub.14}$ toward the first side washer **81** is generated on the first connecting member **2**, both by the fitting between the helical-spline fitting portion **412** of the first side gear **4** and the helical-spline fitting portion **214** of the first connecting member **2**. In addition, during driving, the thrust force $F_{sub.15}$ toward the center washer **80** is generated on the second side gear **5** and the thrust force $F_{sub.16}$ toward the second side washer **82** is generated on the second connecting member **3**, both by the fitting between the helical-spline fitting portion **512** of the second side gear **5** and the helical-spline fitting portion **314** of the second connecting member **3**.

(37) An abutting surface **42c** that faces the center washer **80** in the axial direction and that abuts on

the center washer **80** during driving is formed on a surface on the center washer **80** side of the first wall portion **42** of the first side gear **4**. An abutting surface **52c** that faces the center washer **80** in the axial direction and that abuts on the center washer **80** during driving is formed on a surface on the center washer **80** side of the second wall portion **52** of the second side gear **5**.

(38) During driving, the center washer **80** generates a frictional force between the center washer **80** and the first wall portion **42** of the first side gear **4** by the thrust force $F_{sub.11}$ generated on the first side gear **4** by the meshing between the first side gear **4** and the first pinion gear **61** and the thrust force $F_{sub.13}$ generated on the first side gear **4** by the helical spline coupling between the first side gear **4** and the first connecting member **2**.

(39) During driving, the center washer **80** generates a frictional force between the center washer **80** and the second wall portion **52** of the second side gear **5** by the thrust force $F_{sub.12}$ generated on the second side gear **5** by the meshing between the second side gear **5** and the second pinion gear **62** and the thrust force $F_{sub.15}$ generated on the second side gear **5** by the helical spline coupling between the second side gear **5** and the second connecting member **3**.

(40) During driving, the first side washer **81** generates a frictional force between the first side washer **81** and the first connecting member **2** by the thrust force $F_{sub.14}$ generated on the first connecting member **2** by the helical spline coupling between the first side gear **4** and the first connecting member **2**. During driving, the second side washer **82** generates a frictional force between the second side washer **82** and the second connecting member **3** by the thrust force $F_{sub.16}$ generated on the second connecting member **3** by the helical spline coupling between the second side gear **5** and the second connecting member **3**. The frictional forces generated by these thrust forces $F_{sub.11}$ to $F_{sub.16}$ serve as differential limiting forces that limit differential rotation between the first output shaft **91** and the second output shaft **92** during driving.

(41) On the other hand, during coasting, the thrust force $F_{sub.21}$ toward the first side washer **81** is generated on the first side gear **4** by the meshing between the first side gear **4** and the first pinion gear **61**. The first outer peripheral wall portion **22** of the first connecting member **2** receives this thrust force $F_{sub.21}$ from the end face **41a** of the first cylindrical portion **41**, so that the abutting surface **2a** of the first connecting member **2** abuts on the first side washer **81** and a frictional force is generated. However, the thrust force $F_{sub.23}$ and the thrust force $F_{sub.24}$ that are generated on the first side gear **4** and the first connecting member **2** by the helical spline coupling between the first side gear **4** and the first connecting member **2** cancel each other out due to the end face **41a** of the first cylindrical portion **41** of the first side gear **4** abutting on the end face **22a** of the first outer peripheral wall portion **22** of the first connecting member **2**.

(42) During coasting, the thrust force $F_{sub.22}$ toward the second side washer **82** is generated on the second side gear **5** by the meshing between the second side gear **5** and the second pinion gear **62**. The second outer peripheral wall portion **32** of the second connecting member **3** receives this thrust force $F_{sub.22}$ from the end face **51a** of the second cylindrical portion **51**, so that the abutting surface **3a** of the second connecting member **3** abuts on the second side washer **82** and a frictional force is generated. However, the thrust force $F_{sub.25}$ and the thrust force $F_{sub.26}$ that are generated on the second side gear **5** and the second connecting member **3** by the helical spline coupling between the second side gear **5** and the second connecting member **3** cancel each other out due to the end face **51a** of the second cylindrical portion **51** of the second side gear **5** abutting on the end face **32a** of the second outer peripheral wall portion **32** of the second connecting member **3**.

(43) As described above, during coasting, the thrust force $F_{sub.23}$ and the thrust force $F_{sub.24}$ that are generated by the helical spline coupling between the first side gear **4** and the first connecting member **2** cancel each other out, and the thrust force $F_{sub.25}$ and the thrust force $F_{sub.26}$ that are generated by the helical spline coupling between the second side gear **5** and the second connecting member **3** cancel each other out. Therefore, even when torque that is transferred between the first side gear **4** and the first connecting member **2** and between the second side gear **5** and the second

connecting member **3** is the same between driving and during coasting, the differential limiting forces that are generated during coasting are smaller than the differential limiting forces that are generated during driving. The differential limiting forces that are generated during driving and the differential limiting forces that are generated during coasting are thus different in magnitude.

(44) During driving, the first wall portion **42** of the first side gear **4** is pressed against the center washer **80** by the resultant force of the thrust force $F_{sub.11}$ generated by the meshing between the first side gear **4** and the first pinion gear **61** and the thrust force $F_{sub.13}$ generated by the helical spline coupling between the first side gear **4** and the first connecting member **2**. Therefore, if the contact area between the first side gear **4** and the center washer **80** is small, heat generation and wear tend to occur at the contact surface therebetween. In the present embodiment, however, the first wall portion **42** of the first side gear **4** protrudes radially inward beyond the plurality of straight-spline protrusions **211** of the first connecting member **2**, so that the first wall portion **42** has a large abutting surface **42c** that abuts on the center washer **80**. This reduces heat generation and wear caused by the friction between the first wall portion **42** of the first side gear **4** and the center washer **80**.

(45) During driving, the second wall portion **52** of the second side gear **5** is pressed against the center washer **80** by the resultant force of the thrust force $F_{sub.21}$ generated by the meshing between the second side gear **5** and the second pinion gear **62** and the thrust force $F_{sub.25}$ generated by the helical spline coupling between the second side gear **5** and the second connecting member **3**. In the present embodiment, the second wall portion **52** of the second side gear **5** protrudes radially inward beyond the plurality of straight-spline protrusions **311** of the second connecting member **3**, so that the second wall portion **52** has a large abutting surface **52c** that abuts on the center washer **80**. This reduces heat generation and wear caused by the friction between the second wall portion **52** of the second side gear **5** and the center washer **80**.

(46) The through hole **420** of the first side gear **4** and the through hole **520** of the second side gear **5** function as oil holes through which lubricating oil flows in the differential case **7**. The inner diameter $D_{sub.0}$ of the center washer **80** is larger than the inner diameter $D_{sub.11}$ of the through hole **420** and the inner diameter $D_{sub.21}$ of the through hole **520**, and is large enough not to obstruct the flow of lubricating oil through the through holes **420**, **520**. The through hole **420** of the first side gear **4** and the through hole **520** of the second side gear **5** are also used to remove swarf generated when cutting the plurality of helical-spline protrusions **411**, **511** during manufacturing of the first side gear **4** and the second side gear **5**.

(47) The inner diameter $D_{sub.0}$ of the center washer **80** is smaller than the inner diameter $D_{sub.12}$ of the straight-spline fitting portion **212** of the first connecting member **2** and the inner diameter $D_{sub.22}$ of the straight-spline fitting portion **312** of the second connecting member **3**. This provides large contact areas between each of the first wall portion **42** of the first side gear **4** and the second wall portion **52** of the second side gear **5** and the center washer **80**.

(48) As shown in FIG. 3, $L_{sub.12}$ is larger than $L_{sub.11}$, where $L_{sub.11}$ is the axial length between the end face **21a** of the first tubular base portion **21** and the end face **22a** of the first outer peripheral wall portion **22** in the first connecting member **2**, and $L_{sub.12}$ is the axial length between the end face **41a** of the first cylindrical portion **41** and the first opposing surface **42a** of the first wall portion **42** in the first side gear **4**. Therefore, there is always clearance between the end face **21a** of the first tubular base portion **21** and the first opposing surface **42a** of the first wall portion **42**, so that the first connecting member **2** does not contact the first wall portion **42** of the first side gear **4**. As a result, the end face **21a** of the first tubular base portion **21** and the first opposing surface **42a** of the first wall portion **42** need not be subjected to turning etc., and for example, may be left as forged.

(49) Similarly, $L_{sub.22}$ is larger than $L_{sub.21}$, where $L_{sub.21}$ is the axial length between the end face **31a** of the second tubular base portion **31** and the end face **32a** of the second outer peripheral

wall portion **32** in the third connecting member **3**, and L.sub.22 is the axial length between the end face **51a** of the second cylindrical portion **51** and the second opposing surface **52a** of the second wall portion **52** in the second side gear **5**. Therefore, there is always clearance between the end face **31a** of the second tubular base portion **31** and the second opposing surface **52a** of the second wall portion **52**, so that the second connecting member **3** does not contact the second wall portion **52** of the second side gear **5**. As a result, the end face **31a** of the second tubular base portion **31** and the second opposing surface **52a** of the second wall portion **52** need not be subjected to turning etc., and for example, may be left as forged.

(50) FIG. **4** is a sectional view showing a differential gear **1A** according to a comparative example. In this differential gear **1A**, the first side gear **4** does not have the first wall portion **42**, and the second side gear **5** does not have the second wall portion **52**. However, the first connecting member **2** has a first wall portion **24** extending radially inward from an end on the center washer **80** side of the first tubular base portion **21**, and the second connecting member **3** has a second wall portion **34** extending radially inward from an end on the center washer **80** side of the second tubular base portion **31**. The configuration is otherwise the same as the configuration of the differential gear **1** according to the above embodiment. Therefore, in FIG. **4**, the same components as those of the differential gear **1** are denoted by the same signs as those in FIG. **2**, and repetitive description will be omitted.

(51) With the differential gear **1A** according to the comparative example, the first output shaft **91** can be caused to abut on the first wall portion **24** of the first connecting member **2** and the second output shaft **92** can be caused to abut on the second wall portion **34** of the third connecting member **3** when assembling a vehicle. However, the contact areas between each of the first side gear **4** and the second side gear **5** and the center washer **80** during driving are smaller than in the differential gear **1** according to the embodiment, and heat generation and wear tend to occur at the contact surfaces between each of the first side gear **4** and the second side gear **5** and the center washer **80**. The first wall portion **24** and the second wall portion **34** no longer perform any function once the vehicle is assembled.

(52) In contrast, in the differential gear **1** according to the present embodiment, the first side gear **4** has the first wall portion **42** and the second side gear **5** has the second wall portion **52**. This facilitates positioning of the first output shaft **91** and the second output shaft **92** when assembling the vehicle. Moreover, the first wall portion **42** and the second wall portion **52** can increase the contact areas between each of the first side gear **4** and the second side gear **5** and the center washer **80**. Heat generation and wear can thus be reduced.

(53) In the differential gear **1** according to the present embodiment, the through hole **420** of the first side gear **4** and the through hole **520** of the second side gear **5** function as oil holes through which lubricating oil flows in the differential case **7**. This reduces heat generation and wear of the frictional sliding portions in the differential case **7** due to improved lubricity.

(54) Moreover, in the differential gear **1** according to the embodiment, the first connecting member **2** does not contact the first wall portion **42** of the first side gear **4**, and the second connecting member **3** does not contact the second wall portion **52** of the second side gear **5**. Accordingly, machining of the end face **21a** of the first tubular base portion **21**, the end face **31a** of the second tubular base portion **31**, the first opposing surface **42a** of the first wall portion **42**, and the second opposing surface **52a** of the second wall portion **52** can be omitted, so that machining cost can be reduced.

Additional Notes

(55) Although the present invention is described above based on the embodiment, the embodiment is not intended to limit the invention according to the claims. It should be noted that not all the combinations of features described in the embodiment are essential to means for solving the problem in the invention. The present invention can be modified as appropriate by omitting part of the configurations or adding or replacing configurations without departing from the spirit and scope

of the present invention.

REFERENCE SIGNS LIST

(56) **1** DIFFERENTIAL GEAR **2** FIRST CONNECTING MEMBER **21** FIRST TUBULAR BASE PORTION **211** STRAIGHT-SPLINE PROTRUSION **213** HELICAL-SPLINE PROTRUSION **22** FIRST OUTER PERIPHERAL WALL PORTION **22A** END FACE **3** SECOND CONNECTING MEMBER **31** SECOND TUBULAR BASE PORTION **311** STRAIGHT-SPLINE PROTRUSION **313** HELICAL-SPLINE PROTRUSION **32** SECOND OUTER PERIPHERAL WALL PORTION **32A** END FACE **4** FIRST SIDE GEAR **41** FIRST CYLINDRICAL PORTION **41A** END FACE **411** HELICAL-SPLINE PROTRUSION **42** FIRST WALL PORTION **420** THROUGH HOLE **5** SECOND SIDE GEAR **51** SECOND CYLINDRICAL PORTION **51A** END FACE **511** HELICAL-SPLINE PROTRUSION **52** SECOND WALL PORTION **520** THROUGH HOLE **6** PINION GEAR SET **61** FIRST PINION GEAR **62** SECOND PINION GEAR **7** DIFFERENTIAL CASE **80** CENTER WASHER **91** FIRST OUTPUT SHAFT **92** SECOND OUTPUT SHAFT

Claims

1. A differential gear that distributes a driving force for a vehicle to a first output shaft and a second output shaft, the differential gear comprising: a cylindrical first connecting member including, on an inner periphery of the first connecting member, a plurality of straight-spline protrusions to which the first output shaft is coupled by straight splines; a cylindrical second connecting member including, on an inner periphery of the second connecting member, a plurality of straight-spline protrusions to which the second output shaft is coupled by straight splines; a first side gear combined with the first connecting member; a second side gear combined with the second connecting member; a plurality of pinion gear sets each composed of a first pinion gear and a second pinion gear being meshed with each other, the first pinion gear meshing with the first side gear, and the second pinion gear meshing with the second side gear; a differential case that holds the plurality of pinion gear sets; and a center washer that is disposed between the first side gear and the second side gear, rotation of the center washer relative to the differential case being restricted, wherein the first side gear is formed with: a first cylindrical portion disposed around an outer periphery of the first connecting member and coupled to the first connecting member by helical splines; and a first wall portion extending radially inward from one end on the center washer side of the first cylindrical portion, the first wall portion protruding radially inward beyond the plurality of straight-spline protrusions of the first connecting member, the second side gear is formed with: a second cylindrical portion disposed around an outer periphery of the second connecting member and coupled to the second connecting member by helical splines; and a second wall portion extending radially inward from one end on the center washer side of the second cylindrical portion, the second wall portion protruding radially inward beyond the plurality of straight-spline protrusions of the second connecting member, and during driving in which torque is transferred from the first side gear and the second side gear to the first connecting member and the second connecting member, the center washer generates a frictional force between the center washer and each of the first wall portion of the first side gear and the second wall portion of the second side gear by a thrust force generated by coupling by the helical splines.
2. The differential gear according to claim 1, wherein a through hole extending through the first wall portion in an axial direction is located in a central portion of the first side gear, and a through hole extending through the second wall portion in the axial direction is located in a central portion of the second side gear.
3. The differential gear according to claim 1, wherein the first connecting member includes an annular first outer peripheral wall portion that faces an end face on an opposite side of the first cylindrical portion of the first side gear from the center washer in an axial direction, the second connecting member includes an annular second outer peripheral wall portion that faces an end face

on an opposite side of the second cylindrical portion of the second side gear from the center washer in the axial direction, and during coasting in which torque is transferred from the first connecting member and the second connecting member to the first side gear and the second side gear, the end face of the first cylindrical portion and the first outer peripheral wall portion abut on each other, the end face of the second cylindrical portion and the second outer peripheral wall portion abut on each other, the first connecting member and the first wall portion do not abut on each other, and the second connecting member and the second wall portion do not abut on each other.
